What a Solution Remembers About Its Equation: Groups Are Factors, Terms Interact, and Where Exactness Ends

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Abstract. Given one exact solution $u(t,x)$ of an unknown linear equation with constant coefficients, what does u remember about the equation? We prove that the right unit is not the additive term but the irreducible *factor* of the symbol. (i) u determines its operator P exactly when no partial quotient $(P/P_i)(\partial)u$ is an exponential-polynomial; no finite amount of data can certify this, and a black-box test is impossible, yet it becomes decidable under a modest extra hypothesis. (ii) On a simply connected region u splits into one component per factor of the symbol, uniquely up to an exponential-polynomial overlap of known finite dimension; a split per additive term is in general impossible; the split is stable exactly when the sum of the factors' solution spaces is closed, which for two factors means a positive Friedrichs angle. (iii) Inside a factor, for evolution equations $u_t = \sum_n A_n u$ with commuting terms, the part of the solution due to the interplay of the terms is $\prod_n(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_n \Delta_n$ applied to the datum; it costs $O(N)$ per value but $\Theta(2^N)$ as a table, and it is not continuous in Sobolev norms. (iv) Nonlinear equations such as Burgers admit analogous statements, with sharp asymptotics across a shock. Every statement carries a verification status. Each concept is illustrated by a worked example, and a primer for readers new to ideals and companion matrices is included.

Key words. differential operators, annihilator ideal, factorization, exponential-polynomials, interaction term, Friedrichs angle, Burgers equation

MSC codes. 35E20, 35A25, 13P25, 35Q53

1 Introduction

Consider a quantity that is carried by a flow and diffuses at the same time. A natural model is

$$(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)\,u = 0. \tag{1}$$

Its solutions are sums of a travelling profile and a diffusing one. Suppose you are handed one such solution $u(t,x)$, known exactly on a region of the plane, and are not told the equation. Three questions arise, and this paper answers all three.

- **Q1. Identification.** Can the operator be reconstructed from u, and when is it unique?
- **Q2. Decomposition.** If the operator is built from pieces, can u be split into the parts each piece generates? For (1): is there a travelling part and a diffusing part, and what are they?
- **Q3. Interaction.** If the pieces are the *terms* of the operator (advection, diffusion, ...), what part of u is due to no single term but to their interplay, and can it be computed exactly and cheaply?

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(a) oscillation (b) damping

Figure 1: The idea of the paper on the equation $(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)u = 0$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor of the symbol $(\tau + c\xi)(\tau - \nu\xi^2)$: circles on the line of the transport factor, squares on the parabola of the diffusion factor (schematic). Identifying the equation means finding these two groups.

The setting is deliberately analytic: no noise, no statistical model. The aim is an *exact* statement and a precise account of what is impossible.

The idea in one paragraph In Fourier variables a constant-coefficient operator is a polynomial in (τ, ξ): that of (1) is $(\tau + c\,\xi)(\tau - \nu\,\xi^2)$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor: the modes of the first factor on the line $\lambda = -ic\kappa$, those of the second on the parabola $\lambda = -\nu\kappa^2$ (Fig. 1). The modes therefore fall into groups, and the groups are the *factors*, not the terms $(\partial_t^2,\ c\nu\,\partial_x^3,\ \dots)$ of the expanded operator. Everything that follows makes this precise.

How to read the status labels Each statement carries one of: [proved] (complete proof, independently reviewed), [proved mod. X] (complete except for a cited published theorem X whose source was checked), [conditional] (depends on an unverified lemma), [open], or [false] (a counterexample exists). The reviewers were independent AI agents, not human referees; fifteen core algebraic theorems were additionally machine-checked in Lean 4 (branch directory `proofs/lean`). Grey tags such as (ledger C3-T1) refer to entries of the claim ledger kept with the proofs.

Results at a glance Table 1 lists the main statements; Fig. 2 shows how they depend on each other. Section 2 builds the algebra, Section 3 proves the grouping principle and its stability, Section 4 treats the interaction term, Section 5 asks when a solution determines its equation, Section 6 moves to nonlinear equations, and Section 7 collects the limits. Proofs are in Appendix A; Appendix B is a primer.

2 Derivatives become polynomials

In brief. Before asking what a solution knows, we turn operators into polynomials and solutions into objects that carry an ideal of equations; the one-variable case shows every idea in

Table 1: Main results in the order in which they are proved. "Proved" means a complete proof that survived an independent review.

Topic	Result	Status
Identification	$\mathrm{Ann}(u) = (P) \iff (P/P_i)(\partial)u \notin \mathrm{EP}$ for every irreducible $P_i \mid P$ (Theorem 5.1)	[proved]
Data limits	no finite certificate; black box impossible; decidable with a bound or a holonomic model (Proposition 5.5)	[proved]
Decomposition	$\ker P = \sum_j \ker Q_j$ on simply connected Ω, overlap of dimension $\delta(P)$ (Theorem 3.1)	[proved]
Terms vs. factors	a split per additive term is false in general (Proposition 3.3)	[proved]
Stability	the split is stable iff $\gamma_N > 0$; for two factors, positive Friedrichs angle (Theorem 3.4)	[proved]
Interaction term	$E_{\mathrm{int}} = \prod(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum \Delta_n$; $O(N)$ value, $\Theta(2^N)$ table (Theorem 4.1)	[proved]
Instability	$u_0 \mapsto E_{\mathrm{int}}$ is not continuous in H^m (Proposition 4.2)	[proved]
Nonlinear	Burgers: closed form before the shock, sharp behaviour across it (Section 6)	[proved]/ [proved mod. Kružkov]

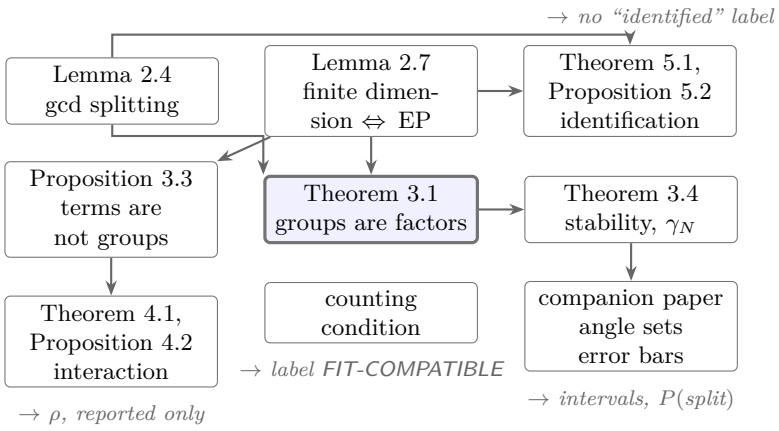


Figure 2: How the results of Section 2 depend on each other (arrows) and where the algorithm uses them (italic). The shaded result is the grouping principle; everything else either prepares it or measures how well it can be carried out on finite data.

miniature.

2.1 Derivatives become polynomials

Throughout, $\Omega \subset \mathbb{R}^2$ is a domain in the variables (t, x) and every operator has constant coefficients,

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials $\psi_{\tau,\xi}(t, x) = e^{\tau t + \xi x}$, $(\tau, \xi) \in \mathbb{C}^2$, are common eigenfunctions of all such operators: $\partial_t \psi_{\tau,\xi} = \tau \psi_{\tau,\xi}$, $\partial_x \psi_{\tau,\xi} = \xi \psi_{\tau,\xi}$ and therefore

$$P(\partial)\, \psi_{\tau,\xi} = P(\tau, \xi)\, \psi_{\tau,\xi}. \tag{2}$$

Differentiation is thus replaced by multiplication, and operators are replaced by polynomials.

Definition 2.1 (Operator ring, symbol, characteristic variety)**.** The constant-coefficient operators form the commutative ring $\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \cong \mathbb{C}[\tau, \xi]$. The polynomial $P(\tau, \xi)$ is the *symbol* of $P(\partial)$, and its zero set $V(P) = \{(\tau, \xi) \in \mathbb{C}^2 : P(\tau, \xi) = 0\}$ is the *characteristic variety*.

By (2), $\psi_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when $(\tau, \xi) \in V(P)$. A Fourier mode in space is the case $\xi = i\kappa$ with wavenumber $\kappa \in \mathbb{R}$; then $P(\tau, i\kappa) = 0$ is the *dispersion relation*, and the roots $\tau = \lambda(\kappa)$ are the temporal rates of the mode $e^{\lambda(\kappa)t + i\kappa x}$.

Convention We keep the algebraic variable ξ in statements about the ring and write $\xi = i\kappa$ only when we evaluate on Fourier modes. With this convention a symbol has real coefficients exactly when the equation is real, and no factor i ever appears inside $\mathcal{R}$.

Example 2.2 (Transport plus diffusion: the running example)*.* The operator $P = (\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)$, with speed $c > 0$ and diffusivity $\nu > 0$, has symbol

$$P(\tau, \xi) = Q_1 Q_2, \qquad Q_1 = \tau + c\,\xi, \qquad Q_2 = \tau - \nu\,\xi^2.$$

On Fourier modes the two factors give two dispersion branches, $\lambda_1(\kappa) = -ic\kappa$ (a wave that travels without decay) and $\lambda_2(\kappa) = -\nu\kappa^2$ (a standing mode that decays). A solution is any superposition of modes on the two branches. The equation is one object; its *two factors* are two groups of modes. Recovering those groups from data is the subject of this paper. We use $c = 1$, $\nu = 0.3$ in all numerical illustrations.

2.2 What a solution knows: the annihilator

A single function satisfies many equations: e^x is annihilated by $\partial_x - 1$, by $\partial_x^2 - 1$ and by every multiple of $\partial_x - 1$.

Definition 2.3 (Annihilator)**.** For a distribution u on Ω, $\mathrm{Ann}(u) = \{Q \in \mathcal{R} :\ Q(\partial)u = 0\}$.

$\mathrm{Ann}(u)$ is an ideal: it is closed under sums and under multiplication by any operator, because $(SQ)(\partial)u = S(\partial)(Q(\partial)u) = 0$. It collects every constant-coefficient equation that u satisfies. We use three classical facts about $\mathbb{C}[\tau, \xi]$ [1]: unique factorization into irreducible polynomials; Bézout's theorem (two polynomials without a common factor have finitely many common zeros); and Hilbert's Nullstellensatz (if q vanishes on the zero set $V(I)$ of an ideal I, then $q^k \in I$ for some k). All three require the complex field, which is why symbols are always considered over $\mathbb{C}$.

Lemma 2.4 (Splitting off the greatest common divisor)**.** *Every nonzero ideal $I \subset \mathcal{R}$ factors as $I = g \cdot J$, where $g = \gcd(I)$ and J is an ideal with finite zero set $V(J)$.*

The proof (Appendix A) is elementary: divide by g, then use Bézout on two elements of J without a common factor. Applied to $I = \operatorname{Ann}(u)$ the lemma gives the decomposition

$$\operatorname{Ann}(u) = G_u\, J_u, \qquad G_u = \gcd \operatorname{Ann}(u), \qquad \Sigma_{\mathrm{p}}(u) = V(J_u) \text{ finite.} \tag{3}$$

The polynomial G_u is the part of the equation that u fills out along whole curves of $\mathbb{C}^2$; the finite set $\Sigma_{\mathrm{p}}(u)$, the *point spectrum* of u, lists the isolated exponentials that u happens to contain. We call G_u the *intrinsic operator* of u: it depends on u alone, whereas an annihilator P supplied from outside may contain spurious factors.

Reading the two pieces The two objects in (3) answer different questions. The polynomial G_u is a differential equation: its zero set $V(G_u)$ is a union of curves in $\mathbb{C}^2$, and each curve is a continuous family of exponential modes $e^{\tau t+\xi x}$, all solving $G_u(\partial)v = 0$. The point spectrum $\Sigma_{\mathrm{p}}(u)$ is a list: the finitely many individual exponential modes that are present in this particular function u and are not accounted for by G_u. In short, G_u identifies an equation that describes a whole family of modes; $\Sigma_{\mathrm{p}}(u)$ identifies the isolated modes that u happens to contain. Example 2.5 shows the two extreme cases.

Example 2.5 (Two extremes). For $u = e^x$, $\operatorname{Ann}(u) = (\tau,\ \xi - 1)$ is the ideal of the single point $(0, 1)$; hence $G_u = 1$ and $\Sigma_{\mathrm{p}}(u) = \{(0, 1)\}$: the function knows one exponential and no equation. For $u = f(x - ct)$ with a profile f that is not a finite sum of exponentials times polynomials, $\operatorname{Ann}(u) = (\tau + c\xi)$; hence $G_u = \tau + c\xi$ and $\Sigma_{\mathrm{p}}(u) = \emptyset$: the function knows its equation completely (Theorem 5.1).

2.3 Exponential-polynomials: the solutions that remember too little

Definition 2.6 (Exponential-polynomials)**.** EP is the set of finite sums $\sum_j p_j(t, x)\, e^{\tau_j t+\xi_j x}$ with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. $\mathrm{EP}(\Omega)$ denotes their restrictions to Ω.

For a function w let $\mathcal{R}w = \operatorname{span}\{\partial_t^j \partial_x^k w :\ j, k \geq 0\}$ be the space of all its derivatives. The map $Q \mapsto Q(\partial)w$ is onto $\mathcal{R}w$ with kernel $\operatorname{Ann}(w)$, so $\mathcal{R}w \cong \mathcal{R}/\operatorname{Ann}(w)$ as vector spaces.

Lemma 2.7 (Finite-dimension criterion)**.** *Let Ω be connected. Then $w \in \mathrm{EP}(\Omega)$ if and only if $\dim_{\mathbb{C}} \mathcal{R}w < \infty$. In that case $V(\operatorname{Ann} w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.*

Proof idea. We show that w is an exponential-polynomial if and only if its derivatives span a finite-dimensional space. ($\Rightarrow$) For one term $w = p(t, x)\, e^{at+bx}$,

$$\partial_t w = (p_t + a\, p)\, e^{at+bx}, \qquad \partial_x w = (p_x + b\, p)\, e^{at+bx}.$$

Differentiation never raises the degree of p, so every derivative of w lies in the finite-dimensional space of polynomials of degree at most $\deg p$ times e^{at+bx}; a finite sum of such terms stays in a finite sum of such spaces. ($\Leftarrow$) If $\mathcal{R}w$ is finite-dimensional, ∂_t and ∂_x map it to itself and act on it as two commuting matrices A and B. Their minimal polynomials m_A and m_B give two constant-coefficient differential equations,

$$m_A(\partial_t)\, w = 0, \qquad m_B(\partial_x)\, w = 0,$$

one in each variable. On every rectangle inside Ω their common solutions are exponential-polynomials, and connectedness extends this representation to all of Ω. The joint eigenvalues (a, b) of A and B are the exponents that occur in w. Details are in Appendix A. □

In brief. Deciding whether $w \in \mathrm{EP}$ is a finite-dimensional linear-algebra question: compute the rank of the derivatives of w. The joint eigenvalues of the two matrices are the exponents, that is, the points of $\mathbb{C}^2$ at which w lives. This is the computational core of everything that follows: the matrix pencil of the companion algorithm paper is exactly this eigenvalue computation, done for ∂_t one wavenumber at a time.

Example 2.8 (Rich and poor solutions of the heat equation). A single heat mode $e^{-\nu\kappa^2 t + i\kappa x}$ and any finite sum of such modes belong to EP. The heat kernel $t^{-1/2}e^{-x^2/(4\nu t)}$ does not: its x-derivatives are polynomials of increasing degree times the same Gaussian, so they span an infinite-dimensional space.

Starting from: one variable x, an ODE $p(D)u = 0$ with constant coefficients, $D = d/dx$, p known. *Why:* in one variable every object of the later sections (minimal annihilator, spectral projectors, companion matrix) appears with explicit formulas and no geometry of curves. *Role:* prototype for Section 5.1 (identification) and Section 3 (the Lagrange projectors reappear as Eq. (11)).

This section introduces, in the simplest setting, objects that reappear in Sections 5.1 and 3: the minimal annihilator, the companion matrix and the spectral projectors.

Remark 2.9 (Setting and symbols).
- We consider an ODE $p(D)u = 0$ in one variable x, with $D = d/dx$ (in this section only; elsewhere D is a diffusion constant).
- The polynomial $p(\lambda) = \prod_j (\lambda - \lambda_j)^{m_j}$ has distinct roots $\lambda_j \in \mathbb{C}$ with multiplicities $m_j \geq 1$, so that $m = \deg p = \sum_j m_j$ is the *order* of the equation (not to be confused with the Sobolev index m of Section 3).
- The set of solutions is the *kernel* $\ker p(D)$, and

$$\ker p(D) = \Big\{ \sum_j q_j(x) e^{\lambda_j x} : \ \deg q_j < m_j \Big\},$$

where each q_j is a polynomial of degree $< m_j$.

Remark 2.10 (Why this form). For one root, $(D - \lambda)\big(q\, e^{\lambda x}\big) = q' e^{\lambda x}$: applying $D - \lambda$ lowers the degree of q by one. Hence $(D - \lambda_j)^{m_j}$ kills $q\, e^{\lambda_j x}$ exactly when $q^{(m_j)} = 0$, i.e. $\deg q < m_j$.

Each root therefore contributes the m_j functions $x^k e^{\lambda_j x}$ $(0 \leq k < m_j)$: these are the natural basis functions of the equation, finitely many, and together they form a basis of $\ker p(D)$, a space of dimension m.

Remark 2.11 (Example). $p(\lambda) = (\lambda - 1)^2(\lambda + 2)^3$ has $\lambda_1 = 1$, $m_1 = 2$, $\lambda_2 = -2$, $m_2 = 3$, order $m = 5$. A basis of the solutions is

$$e^x, \quad xe^x, \quad e^{-2x}, \quad xe^{-2x}, \quad x^2 e^{-2x},$$

so every solution is

$$c_1 e^x + c_2 x e^x + c_3 e^{-2x} + c_4 x e^{-2x} + c_5 x^2 e^{-2x}.$$

2.4 Identification by the Wronskian

Proposition 2.12 ((ODE identification)). *Let $u \in \ker p(D)$. The minimal annihilator of u has degree equal to the number of independent basis functions actually present in u; it equals p if and only if $u, u', \dots, u^{(m-1)}$ are linearly independent (nonvanishing Wronskian), i.e. iff every root of p appears in u with its full multiplicity.*

Reason. $\dim \operatorname{span}\{u, u', u'', \dots\}$ is the degree of the minimal annihilator (Lemma 2.7 in one variable); a missing root lowers it.

2.5 The companion matrix and spectral form

Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$ (monic) and put $X = (u, u', \dots, u^{(m-1)})^\top$. Then $X' = CX$ with

$$C = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ -a_0 & -a_1 & \cdots & -a_{m-1} \end{pmatrix}, \qquad X(x) = e^{xC} X(0).$$

The eigenvalues of C are the roots of p. For distinct roots the matrix exponential has the spectral form

$$\begin{aligned} e^{xC} &= \sum_{j=1}^{m} e^{\lambda_j x}\, \Pi_j, \\ \Pi_j &= \prod_{k \neq j} \frac{C - \lambda_k \mathrm{Id}}{\lambda_j - \lambda_k}, \end{aligned} \tag{4}$$

where the Π_j are the Lagrange projectors onto the eigenspaces. The last row of C carries the whole equation; C is a *matrix over the base field* for an ODE, but, as soon as the coefficients depend on x, or there is a second variable, its last row has entries that are operators and the object becomes a matrix over a noncommutative or polynomial ring. The invariant statement is that the whole equation corresponds to the cyclic module $\mathcal{R}/\operatorname{Ann}(u)$ (Section 2); the companion matrix is a way to compute with it relative to one chosen direction.

Proposition 2.13 ((Projectors of the companion matrix) (ledger II-T2) [proved]). *The projectors Π_j are explicit rational functions of the coefficients, with poles exactly where roots collide; they are polynomial in the coefficients if and only if the corresponding varieties are disjoint. They are* not *orthogonal in general: the companion matrix is not normal for $m = 2$ [false] (the contrary was an early conjecture).*

Hence Hilbert-space orthogonality arguments need an inner product adapted to the problem, not the standard one on the coefficient vector.

2.6 Tensor construction versus module construction

One may encode a PDE in a higher-order tensor of derivatives, $J = \partial_{x_{i_1}} \otimes \cdots \otimes \partial_{x_{i_n}}$, or by the companion form. The comparison, (ledger Gate 3): the *principal symbol tensor* $\sigma_m(P) \in \operatorname{Sym}^m(\mathbb{C}^2)$ only sees the highest-order part of P (for the heat operator it sees ξ^2 alone), so it is an invariant "at infinity"; the *cyclic module* $\mathcal{R}/\operatorname{Ann}(u)$ is the complete invariant. All subsequent results use the module.

Remark 2.14 (Takeaway). This section shows how an ODE's polynomial encodes its solution space. Its roots determine the exponential rates, their multiplicities determine the polynomial prefactors, the companion matrix represents differentiation, and spectral projectors isolate the different modes. The later sections generalize these ideas to multiple commuting derivatives, where the same structure survives but requires more algebraic machinery: the minimal annihilator becomes the ideal $\mathrm{Ann}(u)$ (Section 5.1) and the Lagrange projectors reappear as Eq. (11) (Section 3).

3 Groups are factors, not terms

In brief. The factor decomposition, its stability measured by an angle, and the regularity of the components.

3.1 Groups are factors, not terms

Suppose the operator is known and factored as

$$P = P_1^{e_1} \cdots P_N^{e_N}, \qquad Q_j = P_j^{e_j} \text{ pairwise coprime.} \tag{5}$$

Two ways of splitting a solution suggest themselves: one piece per *factor* Q_j, or one piece per additive *term* of the operator (for $u_t = \nu u_{xx} - c u_x$, one piece for diffusion and one for transport). Only the first works.

Theorem 3.1 (Factor decomposition)**.** *Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and simply connected, and let P be as in* (5)*. Every $u \in \ker P(\partial)$ (in $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$) can be written*

$$u = \sum_{j=1}^{N} u_j, \qquad Q_j(\partial)\, u_j = 0. \tag{6}$$

The tuples $(u_1, \ldots, u_N)$ that sum to zero form a space of exponential-polynomials of dimension exactly

$$\delta(P) = \sum_{i<j} \dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j), \tag{7}$$

the number of intersection points of the varieties $V(Q_i)$, counted pair by pair and with multiplicity.

The number $\dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j)$ therefore has two readings at once. Geometrically it counts the points where the curves $V(Q_i)$ and $V(Q_j)$ meet, with multiplicity. Analytically it is the dimension of the space of exponential-polynomials that solve both Q_i and Q_j, one exponential for each meeting point (times polynomials at a point of higher multiplicity). For two groups these are exactly the functions that can be moved from one group to the other without changing u: each meeting point of the two curves is one shared mode. For three or more groups $\delta(P)$ still gives the dimension of the ambiguity, but the ambiguous tuples need not be sums of modes shared by pairs.

In one variable the statement is Bézout's identity: if $aQ_1 + bQ_2 = 1$, then $u = bQ_2 u + aQ_1 u$ splits u into a solution of Q_1 and a solution of Q_2. In two variables "coprime" no longer gives $1 \in (Q_1, Q_2)$; it only gives finitely many common zeros, and the finite-dimensional quotient

$\mathcal{R}/(Q_1, Q_2)$ measures exactly how much the two groups overlap. The proof combines the exact sequence $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ with the vanishing of the obstruction $\mathrm{Ext}^1(M, C^\infty(\Omega))$ for the finite-dimensional module $M = \mathcal{R}/(Q_1, Q_2)$, which reduces to $H^1_{dR}(\Omega) = 0$ (compare the fundamental principle of Ehrenpreis and Palamodov [4, 12]); see Appendix A.

Example 3.2 (Running example: two groups with a two-dimensional overlap). For $Q_1 = \tau + c\xi$ and $Q_2 = \tau - \nu\xi^2$ the common zeros are $(0, 0)$ and $(c^2/\nu, -c/\nu)$, both simple, so $\delta(P) = 2$. The ambiguity is spanned by the constant function and by $e^{(c^2/\nu)t-(c/\nu)x}$: a constant (or that growing exponential) can be moved from one group to the other without changing u. Among Fourier modes ($\xi = i\kappa$, $\kappa \in \mathbb{R}$) only the first point is visible: the $\kappa = 0$ mode belongs to both groups.

Projectors from the eigenvalues When each factor is an evolution factor, $Q_j = \tau - \Lambda_j(\xi)$ with pairwise distinct branches Λ_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = \Lambda_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - \Lambda_k(\partial_x))$ to $u = \sum_l u_l$ removes every u_l with $l \neq j$ and multiplies u_j by $\prod_{k\neq j}(\Lambda_j - \Lambda_k)(\partial_x)$. On a spatial Fourier mode κ at which no two branches meet this factor is a nonzero number, hence

$$\hat{u}_j = \prod_{k\neq j} \frac{\partial_t - \Lambda_k(i\kappa)}{\Lambda_j(i\kappa) - \Lambda_k(i\kappa)}\ \hat{u}(t, \kappa). \tag{8}$$

This is a multiplier, rational in κ, with poles exactly where two branches meet ($\Lambda_j(i\kappa) = \Lambda_k(i\kappa)$), that is, at the visible part of the overlap counted by $\delta(P)$. Formula (8) is the Lagrange interpolation formula for the spectral projectors of a matrix with distinct eigenvalues, applied to ∂_t restricted to one wavenumber: the *eigenvalues decide the groups*. For the running example, $\hat{u}_1(t, \kappa) = (\partial_t + \nu\kappa^2)\hat{u}/(\nu\kappa^2 - ic\kappa)$, which is singular only at $\kappa = 0$.

Why additive terms do not form groups

Proposition 3.3 (Terms do not give components)**.** *If $P = \mathcal{A}_1 + \mathcal{A}_2$ with constant coefficients, then* $\ker P \cap (\ker \mathcal{A}_1 + \ker \mathcal{A}_2) \subseteq \ker P \cap \ker(\mathcal{A}_1\mathcal{A}_2)$*, and the right-hand side is finite-dimensional whenever $V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$ is finite.*

Proof. If $u = a + b$ with $\mathcal{A}_1 a = 0$ and $\mathcal{A}_2 b = 0$, then $\mathcal{A}_1\mathcal{A}_2 u = \mathcal{A}_2\mathcal{A}_1 a + \mathcal{A}_1\mathcal{A}_2 b = 0$ because the operators commute. At a common zero of P and $\mathcal{A}_1\mathcal{A}_2$ either $\mathcal{A}_1 = 0$, and then $\mathcal{A}_2 = P - \mathcal{A}_1 = 0$, or $\mathcal{A}_2 = 0$ and then $\mathcal{A}_1 = 0$; so $V(P, \mathcal{A}_1\mathcal{A}_2) = V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$. When this set is finite, the ideal $(P, \mathcal{A}_1\mathcal{A}_2)$ has finite codimension, its common kernel consists of exponential-polynomials with exponents in that set (Lemma 2.7), and it is finite-dimensional. □

For the heat equation, $\mathcal{A}_1 = \partial_t$ and $\mathcal{A}_2 = -\nu\partial_x^2$: every sum $f(x) + a(t) + b(t)x$ of a solution of $\mathcal{A}_1$ and one of $\mathcal{A}_2$ is killed by $\mathcal{A}_1\mathcal{A}_2 \sim \partial_t\partial_x^2$, while a heat mode $e^{-\nu\kappa^2 t}\cos\kappa x$ is not. A generic heat solution is *not* "a diffusion part plus a stationary part". The leftover of any term-wise split is the interaction term of Section 4.1.

Where the decomposition fails The hypotheses of Theorem 3.1 are not technical. On a disconnected domain different pieces can satisfy different equations. On a domain with a hole the decomposition can fail: the Laplacian factors as $(\partial_x + i\partial_y)(\partial_x - i\partial_y)$, yet $\log|z|^2$ is

harmonic on an annulus and is not a sum of a holomorphic and an antiholomorphic single-valued function. The obstruction is the first de Rham cohomology of Ω, which is why the algorithm works on rectangles.

3.2 How well separated are the groups? The Friedrichs angle

Theorem 3.1 is algebraic: it says that the split exists, not that it is stable. Stability is a geometric property of the component spaces $M_j = \ker_{H^s} Q_j$, the solutions of each factor in the Sobolev space $H^s(\Omega)$. Let

$$\mathcal{S}:\ \textstyle\bigoplus_{j=1}^{N} M_j \to H^s(\Omega), \qquad \mathcal{S}(u_1,\dots,u_N) = \sum_j u_j,$$

be the *sum operator*, with $\|(u_1,\dots,u_N)\|^2 = \sum_j \|u_j\|_{H^s}^2$, and define the *conditioning constant*

$$\gamma_N = \inf_{0 \neq v \perp \ker \mathcal{S}} \frac{\|\mathcal{S}v\|^2}{\|v\|^2} = \inf \sigma\big(\mathcal{S}^*\mathcal{S}\big|_{(\ker \mathcal{S})^\perp}\big). \tag{9}$$

It is the worst case, over all tuples of pieces, of the size of the sum relative to the size of the pieces. Tuples in $\ker \mathcal{S}$ (the overlap of Theorem 3.1) are excluded. By the triangle and Cauchy–Schwarz inequalities, $\|\sum_j u_j\|^2 \le \big(\sum_j \|u_j\|\big)^2 \le N \sum_j \|u_j\|^2$, hence

$$0 \le \gamma_N \le N.$$

If the spaces are mutually orthogonal, $\|\sum_j u_j\|^2 = \sum_j \|u_j\|^2$ and $\gamma_N = 1$. At the other end, if M_2 approaches M_1 one can take $u_2 \approx -u_1$: the pieces stay of unit size while their sum becomes arbitrarily small, so $\gamma_N \to 0$. A small γ_N is the signature of groups that can almost cancel each other.

For two groups, $\gamma_2 = 1 - c_F$, where the *Friedrichs cosine*

$$c_F = \sup\big\{|\langle a, b\rangle| :\ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\big\} = \cos\theta_F$$

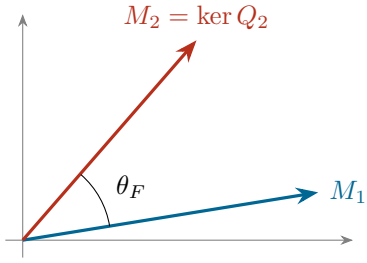
is the cosine of the smallest nonzero angle θ_F between the two spaces (Fig. 3) [2]. In regression terms, M_1 and M_2 are two blocks of regressors and c_F is the largest correlation between a combination of the first block and one of the second: the closer the blocks are to collinear, the less the data can attribute to each.

Theorem 3.4 (Stability of the split)**.** *Every $u \in \ker_{H^s} P$ has components $u_j \in H^s$ if and only if $\sum_j M_j$ is closed and dense in $\ker_{H^s} P$. Closedness is equivalent to $\gamma_N > 0$, and then*

$$\Big(\sum_j \|u_j\|_{H^s}^2\Big)^{1/2} \le \gamma_N^{-1/2}\, \|u\|_{H^s} \tag{10}$$

for the minimum-norm components. If the component spaces are finite-dimensional, as for the sampled kernels used by the algorithm, the sum is always closed and density holds whenever the sampled kernel of P is spanned by those of its factors, so stability reduces to $\gamma_N > 0$.

Density in H^s is a genuine condition on infinite-dimensional spaces. On convex domains solutions of $P(\partial)u = 0$ are limits of exponential solutions, uniformly on compact subsets [10, 8], and every exponential solution lies in $\sum_j M_j$; whether this extends to the H^s norm up to the boundary for general P we do not claim.



$c_F = \cos\theta_F, \quad \gamma_2 = 1 - c_F$
error amplification $\gamma_2^{-1/2}$
running example: $\theta_F \approx 0.60$, $\gamma_2 \approx 0.17$

Figure 3: The Friedrichs angle θ_F between the solution spaces of two factors. A small angle means that large pieces can almost cancel, so recovering the pieces from their sum amplifies errors by up to $\gamma_2^{-1/2} = (1-\cos\theta_F)^{-1/2}$. The values quoted are those measured on the running example in the companion algorithm paper.

Two warnings come with the theorem. First, $\gamma_N > 0$ can fail on infinite-dimensional component spaces: the pieces of a smooth u may then be one derivative rougher than u itself. Second, for $N \geq 3$ groups the pairwise cosines do not determine γ_N: two pairs can be well separated while the three spaces are nearly dependent, so the full Gram operator of $\mathcal{S}$ must be used.

3.3 An explicit projector for evolution factors

For an evolution operator with $P = \prod_{j=1}^{N}(\tau - a_j(\xi))$ and distinct a_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = a_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$ kills every u_l with $l \neq j$ (it contains the factor $\partial_t - a_l$) and multiplies u_j by $\prod_{k\neq j}(a_j - a_k)(\partial_x)$. Hence

$$u_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}\, u, \tag{11}$$

the Lagrange formula of Eq. (4) with C replaced by ∂_t. The result is a Fourier multiplier in x that is rational in ξ, with poles where $a_j(\xi) = a_k(\xi)$; this matches the characterization "projectors rational in ξ iff the factors are coprime" (ledger II-T2/T3).

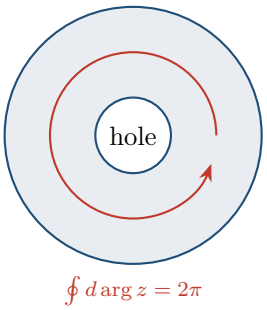
3.4 Where decomposition fails

Proposition 3.5 ((Holes) (ledger C12-P10) [proved]). *On a domain with a hole, the decomposition fails for some P.*

Figure 4 shows the standard obstruction. In the theorem the hypothesis "simply connected" means $H^1_{dR}(\Omega) = 0$, exactly what removes this obstruction; a cohomological criterion for non-simply connected Ω is also available (ledger OP-FUN, C12-Q6).

3.5 Regularity: when are the components in H^m?

The decomposition is an algebraic statement; analysis enters when we want components in a Sobolev space. For two components the governing condition is geometric: the two component spaces M_1 and M_2 must be separated by a positive angle (Figure 5); the next theorem makes this precise for any number of components. Two difficulties appear along the way; they are collected in the box after the theorem.



$u = \log|z-\alpha|^2$ solves $\Delta u = 0$ on the annulus,
but $u \neq h(z) + \overline{g(z)}$ with h, g
single-valued holomorphic: the logarithm
has monodromy (winding number $\neq 0$).

Figure 4: The Laplacian $\partial_x^2 + \partial_y^2 = (\partial_x + i\partial_y)(\partial_x - i\partial_y)$ factors, but the factor decomposition fails on a punctured disc: the obstruction is topological ($H^1_{dR} \neq 0$).

Proposition 3.6 ((Loss of a derivative) (ledger C12-C1) [false])**.** *"If $u \in H^m(\Omega)$ then all $u_j \in H^m$" is false: one derivative can be lost. In other words, the pieces of a smooth function may be rougher than the function itself.*

Theorem 3.7 ((Regularity criterion) (ledger C12-T3, C12-P7) [proved])**.** *All components lie in H^m for every $u \in \ker_{H^m} P$ iff the sum $\sum_j \ker_{H^m} Q_j$ is* closed *and* dense *in $\ker_{H^m} P$. Closedness is equivalent to $\gamma_N > 0$, where*

$$\gamma_N = \inf \sigma\big(S^*S|_{(\ker S)^\perp}\big),$$
$$S : \bigoplus_{j=1}^{N} \ker Q_j \to H^m, \qquad S(u_1, \dots, u_N) = \sum_{j=1}^{N} u_j.$$

For $N = 2$ this is the condition $c_F < 1$ on the Friedrichs cosine. Density holds on every bounded convex Ω, for every P and m (ledger OPF5B-T1) [proved mod. Hörmander [7], Stein].

Remark 3.8 (Reading γ_N). • The space $\bigoplus_j \ker Q_j$ consists of the N-tuples $v = (u_1, \dots, u_N)$ with norm $\|v\|^2 = \sum_j \|u_j\|^2_{H^m}$;
- $Sv = \sum_j u_j$ is their sum;
- S^* is the adjoint of S;
- $(\ker S)^\perp$ removes the tuples that sum to zero (the ambiguity of Theorem 3.1);
- $\inf \sigma$ is the bottom of the spectrum.

Since S^*S restricted to $(\ker S)^\perp$ is self-adjoint and nonnegative, the bottom of its spectrum is the minimum of the Rayleigh quotient

$$\frac{\langle S^*Sv, v\rangle}{\|v\|^2} = \frac{\|Sv\|^2}{\|v\|^2}.$$

Hence the formula of the theorem is equivalent to the more intuitive

$$\boxed{\gamma_N = \inf_{\substack{v \in (\ker S)^\perp \\ v \neq 0}} \frac{\|Sv\|^2_{H^m}}{\|v\|^2}}$$

Remark 3.9 (In words). γ_N is the worst case, over all tuples of pieces, of the ratio between the size of the sum and the size of the pieces.

- If $\gamma_N > 0$ the pieces cannot nearly cancel each other, and

$$\|v\| \le \gamma_N^{-1/2} \|Sv\| :$$

  the pieces are controlled by the sum, so recovering them from u is stable.
- If $\gamma_N \to 0$ there are large pieces whose sum is tiny, and the recovery is unstable.

The case $N = 2$. One finds $\gamma_2 = 1 - c_F$, where

$$c_F = \sup\{|\langle a, b\rangle| : \ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\}$$

is the Friedrichs cosine: the part of v lying in $M_1 \cap M_2$ contributes the ratio 2 and does not affect the infimum, and on the rest

$$\|a + b\|^2 \ge (1 - c_F)(\|a\|^2 + \|b\|^2)$$

with near-equality attained. Hence $c_F < 1 \iff \gamma_2 > 0$, as stated.

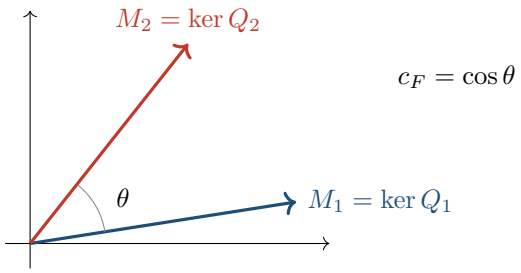


Figure 5: Friedrichs angle θ between two component spaces $M_1 = \ker Q_1$ and $M_2 = \ker Q_2$. A larger angle means a better-conditioned recovery of the pieces from their sum.

For $N = 2$, Figure 5 reads as follows: $c_F = \cos\theta$; if $\theta > 0$ the sum $M_1 + M_2$ is closed and the pieces depend continuously on u; as $\theta \to 0$ the split becomes unstable. For $N \ge 3$ the pairwise angles do not suffice; γ_N, the smallest nonzero eigenvalue of S^*S, is the right conditioning number: it is the conditioning of the linear map "components $\mapsto$ sum", exactly like the smallest nonzero singular value in a least-squares problem.

Remark 3.10 (Warning). [title=Two problems with regularity] **Problem 1: stability [false].** Algebraically the split always exists (Theorem 3.1), but the pieces may be less regular than u. When the angle between the component spaces tends to 0 (that is, $\gamma_N \to 0$) the map "$u \mapsto$ pieces" becomes unstable, and one derivative can be lost (Proposition 3.6). For the algorithm: check $\gamma_N > 0$ before using the components in H^m (Step 5).

Problem 2: three or more components [open]. For $N \ge 3$ the pairwise cosines $c_F(i, j)$ do not decide whether the sum of kernels is closed: an abstract Hilbert-space counterexample exists (ledger S-CHECK-4), and whether it can occur for differential operators is open (ledger C12-Q2).

Remark 3.11 (Why it matters). Algebraically the split always exists (Theorem 3.1); analytically, the components are well behaved exactly when the component spaces are separated by a positive angle (Theorem 3.7). The next section stops asking for a split into factors and asks what survives when one splits by additive terms instead.

4 Inside one group: the interaction term

In brief. What remains when a factor is built from several terms: an exact formula, its cost, its instability, and what happens when the terms do not commute.

4.1 Inside one group: the interaction term and its energy

A single factor may have several additive terms: $Q_2 = \tau - \nu\xi^2$ has one, but the advection–diffusion factor $\tau + c\xi - \nu\xi^2$ has two. Proposition 3.3 says that such a factor cannot be split further into one solution per term. We now measure exactly what is left over.

Consider an evolution $u_t = \mathcal{A}u$ with $\mathcal{A} = \mathcal{A}_1 + \cdots + \mathcal{A}_K$ (the terms), the flow $\Phi(t) = e^{t\mathcal{A}}$ and the flows of the single terms $\Phi_n(t) = e^{t\mathcal{A}_n}$. If term n acted alone it would change the datum by $\Delta_n u_0 = (\Phi_n(t) - \mathrm{Id})u_0$. The *interaction term* is the part of the evolution that is not the sum of these single-term changes,

$$E_{\mathrm{int}}(t) = u(t) - u_0 - \sum_{n=1}^{K} \big(\Phi_n(t)u_0 - u_0\big). \tag{12}$$

Theorem 4.1 (Interaction term)**.** *If the terms commute (in particular, for constant coefficients),*

$$E_{\mathrm{int}}(t) = \Big[\prod_{n=1}^{K}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{K}\Delta_n\Big]u_0 = \sum_{\mathcal{E}\subseteq\{1,\dots,K\},\ |\mathcal{E}|\geq 2} \Big(\prod_{n\in\mathcal{E}}\Delta_n\Big)u_0.$$

For two terms $E_{\mathrm{int}} = \Delta_1\Delta_2 u_0$. On Fourier modes, with symbols $\hat{\mathcal{A}}_n(\kappa)$ of the terms,

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \Big[\prod_{n} e^{t\hat{\mathcal{A}}_n(\kappa)} - 1 - \sum_{n}\big(e^{t\hat{\mathcal{A}}_n(\kappa)} - 1\big)\Big]\hat{u}_0(\kappa). \tag{13}$$

Proof. Commuting terms give $\Phi = \prod_n \Phi_n = \prod_n(\mathrm{Id}+\Delta_n)$. Expanding the product over subsets $\mathcal{E} \subseteq \{1,\dots,K\}$ and removing the subsets with zero or one element gives the formula; on Fourier modes every operator is multiplication by a number. □

Formula (13) costs $O(K)$ operations per wavenumber, although writing out all cross terms would need $2^K - K - 1$ of them. For advection–diffusion, $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t}-1)(e^{-\nu\kappa^2 t}-1)| \leq 2$: the interaction is of the same size as the solution, so it is not a small correction that a regularizer could remove.

Proposition 4.2 (No stable removal)**.** *Let $Q = \tau + c\xi - \nu\xi^2$ (advection–diffusion, terms $-c\partial_x$ and $\nu\partial_x^2$) on a bounded rectangle $\Omega = (0,T)\times(a,b)$. For every $s \geq 0$,*

$$\sup\Big\{\|E_{\mathrm{int}}\|_{L^2(\Omega)}/\|u\|_{H^s(\Omega)} :\ u \in \mathrm{EP},\ Q(\partial)u = 0,\ u \neq 0\Big\} = \infty,$$

so $u \mapsto E_{\mathrm{int}}$ has no continuous extension to $\ker_{H^s} Q$. Moreover, a solution of $Q(\partial)u = 0$ is a sum $v + w$ of solutions of the single-term evolutions $(\partial_t + c\partial_x)v = 0$ and $(\partial_t - \nu\partial_x^2)w = 0$ only if it lies in a finite-dimensional space of exponential-polynomials.

The test solutions are $e^{\lambda_\zeta t+\zeta x}$ with real $\zeta > c/\nu$ and $\lambda_\zeta = \nu\zeta^2 - c\zeta$, legitimate data on a bounded interval: diffusion alone amplifies them by $e^{\nu\zeta^2 t}$, the solution grows only like $e^{\lambda_\zeta t}$, and the interaction must cancel a difference that grows faster than any power of ζ (Appendix A). On a periodic domain with bounded wavenumbers the Fourier multiplier above is bounded by 2; the instability comes from the non-periodic window, which is the situation of real data. The second statement means that a model which writes the data as a sum of single-term pieces, with any penalty on the pieces, leaves a remainder on generic data, and that remainder is the interaction.

Energy We measure the size of a group and of the interaction by L^2 energies on the observation window. The *energy share* of group j and the *interaction ratio* are

$$w_j = \frac{\|u_j\|}{\sum_k \|u_k\|}, \qquad \rho = \frac{\|E_{\text{int}}(T)\|}{\|u(T) - u_0\|}, \tag{14}$$

where T is the length of the time window. The shares w_j say how much of the data each group explains; ρ says how far the evolution inside a multi-term factor is from the superposition of its terms ($\rho = 0$ for a factor with a single term). Interactions occur only *inside* a factor: different factors superpose exactly, which is another way of saying that factors, not terms, are the right groups.

4.2 Definition via the flows of the single terms

Let $S(t) = e^{tA}$ and $S_n(t) = e^{tA_n}$. The contribution that term n *alone* would make to the solution is $S_n(t)u_0 - u_0$. The remainder is the interaction term

$$E_{\text{int}}^{\text{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(S_n(t)u_0 - u_0\big). \tag{15}$$

(The superscript "add" records that this is the *additive* definition; below we write E_{int} for $E_{\text{int}}^{\text{add}}$ unless the distinction matters.)

4.3 Exact formula in the commuting case

Remark 4.3 (Why constant coefficients commute, and what Δ_n is). Constant-coefficient operators are polynomials in ∂_t, ∂_x (Section 2) and the ring $\mathcal{R}$ is commutative, so any two terms commute: $A_1A_2 = A_2A_1$. For commuting operators

$$e^{t(A_1+A_2)} = e^{tA_1}e^{tA_2}, \qquad \text{hence} \qquad S = \prod_n S_n;$$

without commutativity this fails (Section 4.6).

We call

$$\Delta_n = S_n(t) - \mathrm{Id}$$

the *increment* produced by term n alone, where Id is the identity operator and t is fixed; thus $S_n = \mathrm{Id} + \Delta_n$, and $\Delta_n u_0 = S_n(t)u_0 - u_0$ is exactly the contribution of term n in (15). Products of operators are compositions.

Remark 4.4 (Fourier variables). For constant coefficients it is convenient to pass to Fourier variables. With

$$\hat{u}(\kappa) = \int u(x) e^{-i\kappa x}\, \mathrm{d}x,$$

where $\kappa \in \mathbb{R}$ is the frequency:

- the derivative ∂_x becomes multiplication by $i\kappa$;
- each term A_n becomes multiplication by its *symbol* $a_n(\kappa)$ (a polynomial in $i\kappa$);
- e^{tA_n} becomes multiplication by the number $e^{ta_n(\kappa)}$.

Operators turn into numbers and compositions into products of numbers, so for a fixed κ every computation below is scalar.

Theorem 4.5 ((Interaction term) (ledger P1-T2, P1-T3) [proved])**.** *Assume the A_n commute (e.g. constant coefficients). Write $\Delta_n = S_n(t) - \mathrm{Id}$. Then*

$$E_{\mathrm{int}}^{\mathrm{add}} = \Big[\prod_{n=1}^{N} (\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N} \Delta_n \Big] u_0. \tag{16}$$

For $N = 2$ this is simply $E_{\mathrm{int}} = \Delta_1 \Delta_2 u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})u_0$. In Fourier variables, with $\hat{A}_n = a_n(\kappa)$,

$$\widehat{E_{\mathrm{int}}}(t, \kappa) = \Big[\prod_{n} \big(e^{ta_n(\kappa)}\big) - 1 - \sum_{n} \big(e^{ta_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa).$$

Proof. Commutation gives $S = e^{t\sum A_n} = \prod_n S_n$. Substitute $S_n = \mathrm{Id} + \Delta_n$ in (15). For $N = 2$, expand the product:

$$\begin{aligned} S_1 S_2 &= (\mathrm{Id} + \Delta_1)(\mathrm{Id} + \Delta_2) \\ &= \mathrm{Id} + \Delta_1 + \Delta_2 + \Delta_1 \Delta_2, \end{aligned}$$

and $\sum_n (S_n - \mathrm{Id}) = \Delta_1 + \Delta_2$, so the remainder in (15) is $\Delta_1 \Delta_2 u_0$.

For general N, the Δ_n commute, so

$$\prod_{n} (\mathrm{Id} + \Delta_n) = \sum_{S \subseteq \{1, \dots, N\}} \prod_{n \in S} \Delta_n;$$

removing the subsets with $|S| = 0$ and $|S| = 1$ leaves

$$E_{\mathrm{int}} = \sum_{|S| \geq 2} \Delta_S\, u_0, \qquad \Delta_S = \prod_{n \in S} \Delta_n.$$

In words: the interaction is the sum of products of at least two increments, and appears only when at least two terms act together. □

A concrete case. Take advection–diffusion, $u_t = -cu_x + Du_{xx}$, with Gaussian datum $u_0 = g(x; \sigma^2)$, where $g(x; v) = e^{-x^2/2v}/\sqrt{2\pi v}$. The two flows act as a translation and a Gaussian smoothing:

$$S_1 u_0 = g(x - ct; \sigma^2), \qquad S_2 u_0 = g(x; s^2), \qquad u = g(x - ct; s^2), \qquad s^2 = \sigma^2 + 2Dt.$$

Hence

$$E_{\mathrm{int}} = g(x - ct; s^2) - g(x - ct; \sigma^2) - g(x; s^2) + g(x; \sigma^2).$$

Figure 6(a) plots it: it is comparable in size to u itself, so it cannot be discarded as a small correction. In Fourier variables the ratio $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t} - 1)(e^{-D\kappa^2 t} - 1)|$ is at most 2; it vanishes at $\kappa = 0$ and whenever $c\kappa t \in 2\pi\mathbb{Z}$ (the translation returns to the identity), see Figure 6(b).

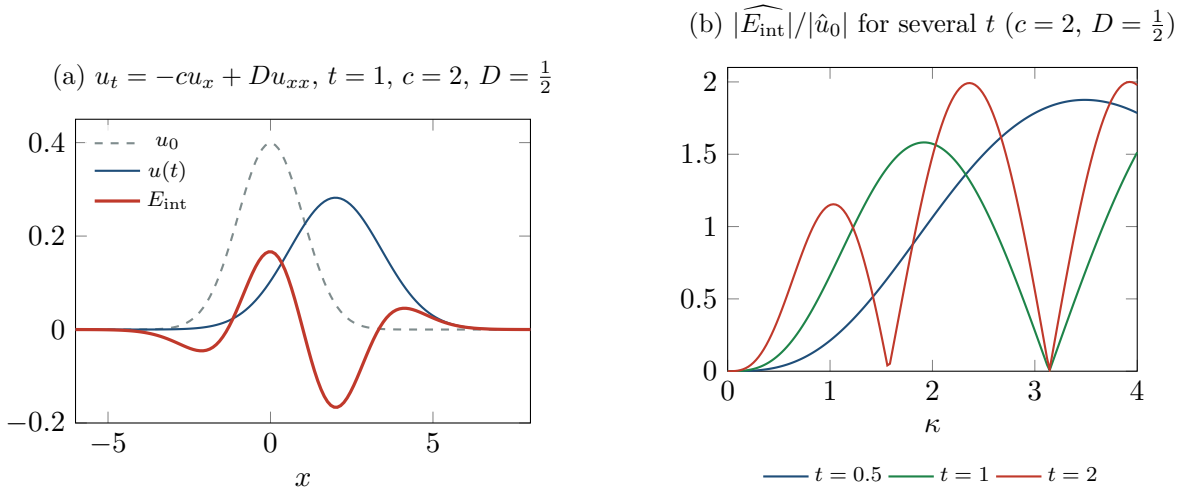


Figure 6: Advection–diffusion with Gaussian datum ($\sigma^2 = 1$, $c = 2$, $D = \frac{1}{2}$). (a) u_0, $u(t)$ and E_{int} at $t = 1$. (b) The Fourier ratio $|\widehat{E_{\text{int}}}|/|\hat{u}_0|$ for three times.

4.4 Cost: $O(N)$ versus 2^N

Expanding (16) in the Δ_n produces one monomial per subset of $\{1, \dots, N\}$ with at least two elements: $2^N - N - 1$ cross terms (Figure 7: the red nodes are the cross terms). Writing the *complete table* of cross terms is therefore exponential. Evaluating a *value* is cheap, because we never expand the product. The algorithm uses exactly the product (Fourier) form of Theorem 4.5,

$$\widehat{E_{\text{int}}}(t, \kappa) = \Big[\prod_{n=1}^{N} e^{t a_n(\kappa)} - 1 - \sum_{n=1}^{N} \big(e^{t a_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa),$$

which for a fixed κ needs the N numbers $e^{t a_n(\kappa)}$, $N - 1$ products and N subtractions: $O(N)$ operations once $\hat{u}_0(\kappa)$ is known.

Remark 4.6 (Two remarks on cost). **What "value" means.** Here "value" means one Fourier value; recovering $E_{\text{int}}(t, x)$ at a point of space needs an inverse Fourier transform, whose cost is not counted in the $O(N)$.

Where 2^N comes from. The count $2^N - N - 1$ refers only to writing out all cross terms one by one; the product form never does so (Step 6 of the algorithm).

Proposition 4.7 ((Complexity) (ledger P1-T3) [proved]). *In the commuting exponential-type class, a value of E_{int} costs $O(M(N + \sum d))$ for M evaluation points; aggregated by order $O(MN^2)$; the full table $\Theta(M2^N(L + 1)^2)$.*

4.5 Time-dependent perturbation theory

Remark 4.8 (Why this subsection). The product formula (16) needs commuting terms. When terms do not commute, or when boundary conditions are imposed, there is no closed formula, but E_{int} is still defined by (15). Perturbation theory expands the flow in powers of the interaction. From this expansion one reads

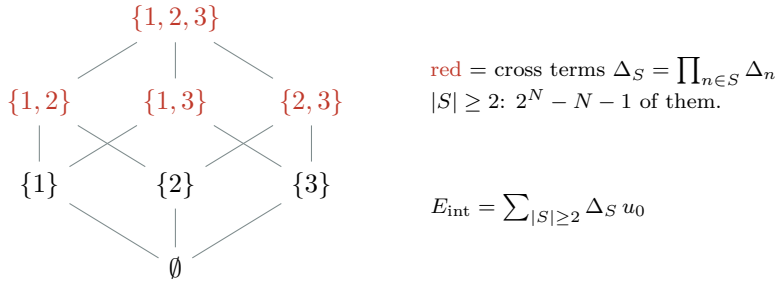


Figure 7: Subset lattice for $N = 3$ terms: one node per subset S of terms, one monomial Δ_S per node with $|S| \geq 2$. The count $2^N - N - 1$ is what makes the full table exponential.

(i) that E_{int} is well defined for all times (convergence),
(ii) approximations obtained by truncating the series order by order, and
(iii) the size of E_{int} for small t.

Split $A = A_1 + B$, where A_1 is the principal term (for instance diffusion) and B, the remaining terms, is treated as a perturbation of A_1. The Dyson–Phillips series is

$$\begin{aligned} S(t) &= \sum_{k\geq 0} S^{(k)}(t), \\ S^{(0)}(t) &= e^{tA_1}, \\ S^{(k+1)}(t) &= \int_0^t e^{(t-s)A_1}\, B\, S^{(k)}(s)\, \mathrm{d}s, \end{aligned} \tag{17}$$

the same expansion that gives the interaction picture of quantum mechanics [3].

Remark 4.9 (Reading the series). • $S^{(0)} = e^{tA_1}$ is the flow of A_1 alone;
• $S^{(k)}$ collects the contributions in which B acts k times, at intermediate times $s \in (0, t)$, with the flow of A_1 in between.

The terms with $k \geq 1$ are therefore the part of the flow in which B acts on the evolution generated by A_1, that is, the interaction. (When A_1 and B commute, the first-order term is

$$\int_0^t e^{(t-s)A_1} B e^{sA_1}\, \mathrm{d}s = t\, B e^{tA_1},$$

consistent with Section 4.)

What it gives. The series converges for all t when A_1 generates an even-order dissipative semigroup (e.g. diffusion) and B is a lower-order L^∞ perturbation, also with Dirichlet conditions (ledger P3-T1) [proved mod. Miyadera–Voigt [5]]: so E_{int} is well defined and can be approximated by truncation even without a closed formula. Each $S^{(k)}$ is the order-k part of the interaction. For small times, for boundary-induced interaction, with k the order of contact of the datum with the boundary, $\|E_{\text{int}}\| \sim t^{k/2+3/4}$ (ledger P3-T3, OP-F5A) [proved]: the smoother the datum is against the boundary (larger k), the faster the interaction becomes negligible as $t \to 0$.

4.6 Non-commuting terms

If $[A_1, A_2] \neq 0$, then $e^{t(A_1+A_2)} \neq e^{tA_1} e^{tA_2}$; by Baker–Campbell–Hausdorff,

$$e^{tA_1} e^{tA_2} = \exp\Big(t(A_1 + A_2) + \tfrac{t^2}{2}[A_1, A_2] + O(t^3)\Big).$$

Proposition 4.10 ((Lie-algebra dichotomy) (ledger P1-T4, P1-T5) [proved]). *If the Lie algebra generated by the terms is solvable, an exact formula with finitely many scalar functions exists (Wei–Norman). Otherwise only a local expansion exists, and* ***no finite product of the flows of the individual terms*** *reproduces* $e^{t(X+\varepsilon Y)}$.

Remark 4.11 (Why it matters). For commuting terms the interaction is explicit and cheap to evaluate, but it cannot be made to vanish nor be computed stably in H^m; for non-commuting terms even the explicit form is lost in general. These are the linear-case limits against which the nonlinear extension of Section 6 must be read.

5 When does a solution determine its equation?

In brief. The identification criterion, uniqueness, and what finite data can and cannot certify. This explains why the algorithm of the companion paper never reports "identified".

5.1 When does a solution determine its equation?

Let Ω be open and connected (on a disconnected set the pieces may obey different equations), let $u \in \mathcal{D}'(\Omega)$ satisfy $P(\partial)u = 0$, and let $P = \prod_i P_i^{e_i}$ (up to a nonzero constant factor) be the factorization of P into pairwise non-associate irreducible polynomials P_i with multiplicities e_i. The question is whether $\mathrm{Ann}(u) = (P)$: then every constant-coefficient equation satisfied by u is a consequence of P, and P is determined by u.

Theorem 5.1 (Identification criterion). *Put* $v_i = (P/P_i)(\partial)\, u$ *and* $v_i' = (P/P_i^{e_i})(\partial)\, u$.

(a) $\mathrm{Ann}(u) = (P)$ *if and only if* $v_i \notin \mathrm{EP}(\Omega)$ *for every* i.

(b) $V(\mathrm{Ann}\, u) = V(P)$ *if and only if* $v_i' \notin \mathrm{EP}(\Omega)$ *for every* i.

(c) If P *is square-free, (a) and (b) coincide. If* P *is monic in* τ *of degree* m, *then* $\mathrm{Ann}(u) = (P)$ *if and only if the only* $(\varphi_0, \dots, \varphi_{m-1}) \in \mathbb{C}[\xi]^m$ *with* $\sum_{j<m} \varphi_j(\partial_x)\, \partial_t^j u = 0$ *is zero.*

The idea is to strip one irreducible factor and look at what is left. The function v_i solves $P_i(\partial)v_i = 0$. If v_i is a generic solution of that factor, no smaller operator kills u. If v_i is an exponential-polynomial, some h coprime to P_i kills it, and then $h \cdot P/P_i$ is an annihilator of u that is not a multiple of P. The converse uses Lemma 2.4; the full proof is in Appendix A.

Proposition 5.2 (Uniqueness and the intrinsic operator). *(a) If* $\mathrm{Ann}(u) = (P)$, *then* P *is determined by* u *up to a nonzero constant and is the unique annihilator of minimal degree.*

(b) $\mathrm{Ann}(u) = (P)$ *if and only if* $\Sigma_{\mathrm{p}}(u) = \emptyset$ *and* $G_u \sim P$ *(equality up to a constant).*

(c) For any annihilator P, *the multiplicity of* P_i *in* G_u *equals* $e_i - \max\{k \le e_i : (P/P_i^k)(\partial)u \in \mathrm{EP}\}$. *Hence* G_u *is computed from any annihilator by at most* $\sum_i e_i$ *membership tests "*$\in \mathrm{EP}$*".*

Example 5.3 (The heat equation, continued). For $P = \tau - \nu\xi^2$ (irreducible) the criterion reads: $\mathrm{Ann}(u) = (P)$ if and only if $u \notin \mathrm{EP}$. One mode $e^{-\nu\kappa^2 t + i\kappa x}$ has $\mathrm{Ann}(u) = (\tau + \nu\kappa^2,\ \xi - i\kappa)$, a point of $V(P)$; it cannot distinguish the heat equation from the pure decay $\partial_t + \nu\kappa^2$, or from any other equation whose variety passes through the point $(-\nu\kappa^2, i\kappa)$. The heat kernel is not in EP, so every constant-coefficient equation it satisfies is a multiple of $\partial_t - \nu\partial_x^2$.

Remark 5.4 (Curves and points). The two cases of Example 5.3 sit at the two ends of the decomposition (3). A finite sum of modes $u \in \mathrm{EP}$ has an annihilator with finitely many zeros, so

$$u \in \mathrm{EP}: \qquad G_u = 1, \qquad \Sigma_{\mathrm{p}}(u) = \{(\tau_j, \xi_j)\} \text{ finite (its exponents)},$$

whereas the zero set of a nonconstant polynomial in two variables is a whole algebraic curve. For the heat kernel the situation is reversed:

$$u = t^{-1/2} e^{-x^2/(4\nu t)}: \qquad G_u = \tau - \nu\xi^2, \quad V(G_u) \text{ a curve}, \qquad \Sigma_{\mathrm{p}}(u) = \emptyset.$$

By Proposition 5.2(b) it is exactly this pair, an empty point spectrum together with $G_u \sim P$, that gives $\mathrm{Ann}(u) = (P)$: the solution fills the whole curve of its equation and contains no isolated mode besides.

5.2 What finite data can and cannot certify

Theorem 5.1 speaks about u, not about finitely many numbers. Two facts frame what an algorithm can promise.

Recall that $\mathcal{R}u = \mathrm{span}\{\partial_t^j \partial_x^k u : \ j, k \geq 0\}$ is the vector space generated by the derivatives of u (Section 2.3). The *jet* of order M of u at a point $z_0 = (t_0, x_0)$ is the list of its derivatives of order at most M at that point; for example

$$j^2_{z_0}(u) = (u, \ u_t, \ u_x, \ u_{tt}, \ u_{tx}, \ u_{xx})\big|_{(t,x)=z_0}.$$

Proposition 5.5 (Finite data)**.** *Let Ω be connected and $z_0 \in \Omega$.*

(i) If $\dim_{\mathbb{C}} \mathcal{R}u \leq n$, the jet of order $2n-1$ of u at z_0 determines $\mathrm{Ann}(u)$ among all functions u' with $\dim_{\mathbb{C}} \mathcal{R}u' \leq n$.

(ii) For every u and every M there is a polynomial, hence an exponential-polynomial, with the same jet of order M at z_0; and for finitely many sample points there is an exponential-polynomial with the same values. Consequently, without a bound on $\dim \mathcal{R}u$, no finite set of jets or samples proves $\mathrm{Ann}(u) = (P)$.

Proof. (ii) The Taylor polynomial of order M at z_0 has the same jet, and polynomial interpolation in two variables matches any finite set of values. Both are exponential-polynomials, and an exponential-polynomial never has $\mathrm{Ann} = (P)$ when P has an irreducible factor (Theorem 5.1: all v_i are then exponential-polynomials). (i) The space $\mathcal{R}u$ is invariant under differentiation and, by Lemma 2.7, consists of real-analytic exponential-polynomials. Let K_M be the subspace of elements of $\mathcal{R}u$ whose jet of order M vanishes at z_0. Then $K_{M+1} = \{f \in K_0 : \ \partial_t f, \partial_x f \in K_M\}$, so once $K_{M+1} = K_M$ the chain is constant, and a constant K_M is a derivative-invariant space of functions vanishing at z_0 with all their derivatives, hence $\{0\}$. Since $\dim K_0 \leq n-1$, the chain reaches $\{0\}$ by $M = n-1$: the jet of order $n-1$ at z_0 is injective on $\mathcal{R}u$, and likewise on $\mathcal{R}u'$. For Q of degree at most n, $Q \in \mathrm{Ann}(u)$ iff the jet of order $n-1$ of $Q(\partial)u$ vanishes at z_0, a condition on the jet of order $2n-1$ of u; the same holds for u'. Finally $\mathrm{Ann}(u)$ has codimension at most n, so a Gröbner basis for a degree-compatible order consists of polynomials of degree at most n; the same holds for $\mathrm{Ann}(u')$, and two ideals generated in degree at most n that contain the same polynomials of degree at most n are equal. □

The consequence for learning is a design rule: without an a priori bound on $\dim \mathcal{R}u$, which data never supply, an algorithm can refute a candidate equation, or report that it is compatible with the data at the resolution available, but it cannot certify it. Its output must carry a label that says which of these happened.

6 Nonlinear equations

In brief. The same two questions, asked of Burgers-type equations: the interaction term for the flows of single terms, behaviour before, at and after a shock, and identification relative to a dictionary.

Starting from: a scalar equation in one space variable, $u_t = \sum_n T_n(u)$, each T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$; Ω bounded and simply connected; the exact solution is known. *Why:* to test which of the notions of Sections 5.1–4 survive when the terms are nonlinear. *Role:* extension of the chain. *Result:* weaker, relative statements (exact expansion of E_{int}, sharp asymptotics across a shock for Burgers, identification relative to a dictionary). Benchmarks: Burgers, KdV–Burgers, Fisher–KPP; KdV and NLS only as controls.

We consider $u_t = \sum_n T_n(u)$ with each T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$. Benchmarks: Burgers $u_t = -uu_x + \nu u_{xx}$, KdV–Burgers, Fisher–KPP; integrable cases (KdV, NLS) serve as controls.

6.1 Interaction term for flows of single terms

With S_n the (nonlinear) flow of $u_t = T_n(u)$ alone, (15) still defines $E_{\text{int}}^{\text{add}}$. Its Taylor expansion in t:

Theorem 6.1 ((Jet expansion) (ledger S4T-T1) [proved]). $E_{\text{int}}^{\text{add}} = \dfrac{t^2}{2} \displaystyle\sum_{n \neq m} T_n'[\, T_m(u_0)\,] + O(t^3)$,

where $T_n'[v]$ is the Fréchet derivative of T_n at u_0 in the direction v.

Example 6.2 (Burgers). $T_1 = -uu_x$, $T_2 = \nu u_{xx}$. Then $T_1'[v] = -(vu_x + uv_x)$ and $T_2'[v] = \nu v_{xx}$, so

$$T_1'[T_2 u_0] = -\nu\,(u_0'' u_0' + u_0 u_0'''),$$
$$T_2'[T_1 u_0] = -\nu\,(3u_0' u_0'' + u_0 u_0'''),$$

and half the sum times t^2 gives

$$\begin{aligned} E_{\text{int}} &= \tfrac{1}{2}(T_1'[T_2 u_0] + T_2'[T_1 u_0])t^2 + O(t^3) \\ &= -\nu(u_0 u_0''' + 2u_0' u_0'')t^2 + O(t^3). \end{aligned}$$

6.2 Before, at, and after the shock

The interaction term of Burgers changes qualitatively with time, and each regime calls for different tools:

Before shock formation smoothness permits perturbative expansions and derivative-based estimates.

At breaking the inviscid gradient blows up, invalidating estimates that require smoothness.

After shock formation pointwise behavior can be discontinuous, so L^1 estimates become more informative than pointwise bounds.

Figure 8 recalls why a shock appears at all in the flow of the single term $-uu_x$: along each characteristic u is constant and the characteristic moves with speed u, so faster parts overtake slower ones. For $u_0 = \sin x$ the characteristics are

$$x = \xi + t \sin \xi$$

(the parameter s in the figure plays the role of the time t); since

$$\frac{\partial x}{\partial \xi} = 1 + t\cos\xi$$

vanishes first at $\xi = \pi$, the breaking time is $t^* = 1$.

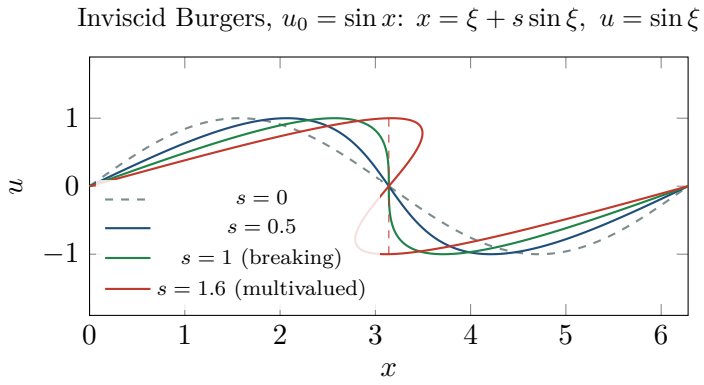


Figure 8: Characteristic solution of the inviscid flow of $-uu_x$. At $s = 1$ the profile has a vertical tangent: the gradient blows up. Beyond that the curve folds; the physical (entropy) solution replaces the fold by a jump at $x = \pi$ (equal-area rule), shown dashed. This breaking is what makes E_{int} discontinuous after the shock.

Before the shock ($t < t^*$): smooth regime. Smoothness permits perturbative expansions and derivative-based estimates.

- E_{int} has a closed form at first order in ν, with remainder $O(\nu^2\varepsilon^2 t^3)$ (ledger S4T-T2) [proved]; it blows up at the breaking time t^* (ledger S4T-P3). (This is the same family of expansions as Theorem 6.1 and Example 6.2.)
- The Cole–Hopf substitution $u = -2\nu\varphi_x/\varphi$ turns Burgers into the heat equation $\varphi_t = \nu\varphi_{xx}$; the inverse scattering transform linearizes KdV and NLS. These are used as the "exact linearization" of the nonlinear solution [6].

At breaking ($t = t^*$): estimates that need smoothness fail. The inviscid gradient u_x blows up (vertical tangent, $s = 1$ in Figure 8), so every statement that needs bounds on $u_x, u_{xx}, \ldots$ stops being usable at t^*: the closed form of E_{int} blows up, and estimates that require smoothness, such as those behind the jet expansion of Theorem 6.1, are no longer valid.

After the shock ($t > t^*$): discontinuous regime. The flow of $-uu_x$ alone forms a shock, so E_{int} jumps by amounts independent of ν; this is forced by the definition. A function with a jump is not in H^1 in one dimension, so the Sobolev norms of Sections 3 and 4 are no longer the natural ones: pointwise values are unreliable, while the *measure* of the discrepancy is well

behaved. The meaningful quantities are therefore L^1 sizes, collected below:

$$\|E_{\mathrm{int}}\|_{L^1} = \Theta(\nu), \tag{18}$$

$$\sup_{t\le T} \|E_{\mathrm{int}}\|_{L^1} = \nu|\log\nu| + O(\nu), \tag{19}$$

$$\|S - S_1S_2\|_{L^1} = \Theta(\sqrt{\nu}). \tag{20}$$

The first line holds at fixed t, each shock contributing $4\nu\log(2\cosh(\rho/2))$; the second has exact leading coefficient 1. Sources: the first line is (ledger OPA-T1) [proved mod. Kružkov uniqueness [9]]; the second is (ledger OPF5A-T1/T2); the third is (ledger OPF5D-T6) [proved]. The second line holds for data with a single birth of shock such as $\sin x$, and is conditional on a localization lemma in general [conditional].

Regime	What happens	Tools and results
Before the shock, $t < t^*$	the profile is smooth, characteristics do not cross	jet expansion (Thm. 6.1), first-order closed form of E_{int}, Cole–Hopf
At breaking, $t = t^*$	the inviscid gradient blows up (vertical tangent)	smoothness-based estimates fail; E_{int} blows up
After the shock, $t > t^*$	the single-term flow is discontinuous (equal-area rule)	L^1 estimates: $\Theta(\nu)$, $\nu\|\log\nu\|$, $\Theta(\sqrt{\nu})$

Table 2: The three regimes of the Burgers interaction term and the tools that apply in each.

6.3 Identification relative to a dictionary

Nonlinear terms cannot be handled by the ideal $\mathrm{Ann}(u)$ of Section 2; instead one fixes a finite candidate list. Choose a dictionary $\mathcal{D}$ of monomials of degree $\le d$ in the jet, of order $\le k$, and ask which polynomial relations $F(u, u_x, \dots) = 0$ the data satisfy. Notation: V is the span of $\mathcal{D}$, $I(u)$ the ideal of all differential relations satisfied by u, $[F]$ the differential ideal generated by F, and $\mathcal{J}_u$ the jet-level obstruction (a different object from the cofactor J_u of Section 2, hence the calligraphic letter).

Theorem 6.3 ((Dictionary criterion) (ledger S4ID-T1) [proved]). *For smooth u on any Ω there is an exact sequence $0 \to [F]\cap V \to I(u)\cap V \to \mathcal{J}_u \cap \psi(V) \to 0$. In particular*

$$I(u)\cap V = [F]\cap V \iff \mathcal{J}_u \cap \psi(V) = 0.$$

The correct statement is therefore "the relations are exactly F *modulo its differential consequences*"; the literal "$I(u)\cap V = \langle F\rangle$" is false in complete dictionaries (they contain $\partial_x F$, etc.) (ledger S4ID-P1). Concrete criteria: for Burgers with N shocks, u identifies the equation in the evolutionary dictionary iff $d \le N$ (ledger S4ID-T2, OP-T7); for N-soliton KdV the identifying polynomial has total degree $(N+2)2^{N-1}$ (generic $N = 2$: 8) (ledger OP-K15) [proved mod. Novikov 1974 [11]]. Finally, "closed form $\Rightarrow$ does not identify" is itself a theorem (ledger S4ID-T4).

6.4 Linear versus nonlinear: the same idea, two structures

Sections 2–4 and the present one answer the same question: *what does a single solution u remember about its equation?* In both settings we collect all relations satisfied by u and ask

	Linear approach (Sections 2–4)	Nonlinear approach (this section)
Space of candidate equations	the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$ of constant-coefficient operators	a finite *dictionary* $\mathcal{D}$ of jet monomials of degree $\leq d$ and order $\leq k$; $V = \operatorname{span} \mathcal{D}$
Relations satisfied by u	$\operatorname{Ann}(u)$, an ideal of $\mathcal{R}$	$I(u)$, the ideal of all differential relations
What the equation generates	the principal ideal (P)	the differential ideal $[F]$
Criterion for identification	$\operatorname{Ann}(u) = (P)$ (Theorem 5.1)	$I(u) \cap V = [F] \cap V$, i.e. F *modulo its differential consequences* (Theorem 6.3)
What escapes the equation	the cofactor J_u: the finitely many exponentials in u ($\sigma_p(u) = V(J_u)$)	the jet-level obstruction $\mathcal{J}_u \cap \psi(V)$ (exact sequence of Theorem 6.3)
Solutions that do not identify	$u \in \mathrm{EP}$	closed-form solutions ((ledger S4ID-T4))
Concrete criteria	finite linear algebra (Lemma 2.7)	Burgers with N shocks identifies iff $d \leq N$; N-soliton KdV: degree $(N+2)2^{N-1}$

Table 3: The linear and nonlinear identification problems side by side.

when these relations are generated by the equation we were given. What changes is the algebraic structure in which the relations live.

Remark 6.4 (Where the parallel breaks). **Linear case.** The solutions form a vector space and the annihilators an ideal: one can add, multiply by operators and factor, which is why the Nullstellensatz applies and the test is a finite-dimensional computation.

Nonlinear case. There is no linear structure on the solutions: one must fix a finite space V of candidates in advance, and the identification is *relative to* V, never absolute. The literal statement

$$I(u) \cap V = \langle F \rangle$$

is false in complete dictionaries, which contain $\partial_x F$, so the correct statement is "modulo differential consequences".

Remark 6.5 (Takeaway). Both approaches ask what one solution determines about its equation. The linear one answers with the annihilator ideal $\operatorname{Ann}(u)$ and its generator G_u; the nonlinear one answers with a dictionary and the differential ideal of relations, and only up to differential consequences. In the algorithm of the companion paper Steps 1–6 use the structure of $\operatorname{Ann}(u)$, while Step 7 switches to the dictionary and returns a relative identification.

7 Limits, open problems, and status

Starting from: no new assumptions. *Why:* to state what is impossible, what is open, and how reliable the results are. *Role:* delimits the claims of the paper.

What is impossible (all [proved]).

- finite certificates of identification, and black-box identification (Section 5.2);
- per-term decomposition of the solution (Proposition 3.3);

Table 4: The kinds of claim made in the paper.

Kind	Statements
proved	groups are factors with overlap $\delta(P)$ (Theorem 3.1); terms are not groups (Propositions 3.3 and 4.2); identification criterion and its limits on finite data (Theorem 5.1 and Propositions 5.2 and 5.5); stability via γ_N (Theorem 3.4); Bayesian error bars via the angle (the companion algorithm paper); counting condition
design rule	labels instead of "identified"; distributions of ρ instead of values; decisions on the raw observations; growth of the class instead of coefficient penalties; minimum-norm split of the overlap
measured	0/20 confident wrong answers on random equations not used in development; 60/60 factor sets on grid designs with independent fields and noise; 30/30 on the development benchmark; coefficient coverage 0.94 at nominal 0.90

- stable regularization of E_{int} (Proposition 4.2);
- identification by restricting variable-coefficient operators to one solution;
- decomposition on domains with holes (Figure 4);
- a finite product of single-term flows reproducing non-commuting dynamics.

Open or conditional.

- identification for analytic KdV data with nonzero reflection coefficient (ledger INT-Q1) [open];
- time-uniform shock estimate with several shock births (ledger OPA-Q1) [conditional];
- density of the sum of kernels on nonconvex domains [open];
- exact values $\delta_N(K)$ of an invariant for general N (ledger OPF5C-Q1) [open];
- nonlinear asymptotics with derivatives (ledger S4ID-P8) [conditional];
- closedness of the sum of kernels for $N \geq 3$ differential factors (ledger C12-Q2) [open].

Honest assessment. Many ingredients are classical (Malgrange–Ehrenpreis, Nullstellensatz, Cole–Hopf, inverse scattering). The contribution is the *package*: sharp characterizations, negative theorems, and counterexamples, each with a verified status. Independent reviewers were AI agents. Fifteen theorems of the algebraic core (the E_{int} identity for commuting flows, the comaximal factor decomposition with explicit projector, companion normality for $k = 2$, irreducibility results) are formalized in Lean 4/Mathlib without `sorry`; the analytic parts (distributions, Sobolev spaces, semigroups) are not, since Mathlib lacks the libraries. Several citations were verified only on secondary sources; a result relied upon should be traced to its primary source. Out of scope so far: a variable potential treated as noise (Schrödinger/Dirac), matrix-valued coefficients (4×4 Dirac), and any statistical preprocessing.

A Proofs

Throughout, $\Omega \subset \mathbb{R}^2$ is open and connected, $\mathcal{A}$ is $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $\mathcal{R} = \mathbb{C}[\tau, \xi]$ acts on $\mathcal{A}$ by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. For $z = (\tau_z, \xi_z) \in \mathbb{C}^2$, $\mathfrak{m}_z = (\tau - \tau_z, \xi - \xi_z)$ is its maximal ideal and $\psi_z(t, x) = e^{\tau_z t + \xi_z x}$.

A.1 Algebraic toolkit

Proof of Lemma 2.4. Coprime polynomials meet in finitely many points. Let $f, g \neq 0$ be coprime. By Gauss's lemma they are coprime in the principal ideal domain $\mathbb{C}(\xi)[\tau]$, so $af + bg = 1$ there; clearing denominators, $a'f + b'g = r(\xi) \neq 0$ with $a', b' \in \mathcal{R}$. At a common zero $r(\xi_0) = 0$, which leaves finitely many ξ_0; exchanging the variables, finitely many τ_0. Hence $V(f, g)$ is finite.

Splitting. Put $J = \{f/g : \ f \in I\}$ with $g = \gcd(I)$; it is an ideal and $I = gJ$, and no prime divides all of J. If $J \neq \mathcal{R}$, pick $0 \neq f \in J$ with distinct prime factors $\pi_1, \dots, \pi_r$ and, for each a, $g_a \in J$ with $\pi_a \nmid g_a$. The sets $\{c \in \mathbb{C}^r : \pi_a \mid \sum_b c_b g_b\}$ are proper linear subspaces, so some c avoids all of them; then f and $\sum_b c_b g_b \in J$ are coprime and $V(J) \subseteq V(f, \sum_b c_b g_b)$ is finite. By the Nullstellensatz a power of $\sqrt{J}$ lies in J, and $J = \mathcal{R}$ when $V(J) = \emptyset$. □

Proof of Lemma 2.7. ($\Rightarrow$) $Q(\partial)(q\,\psi_z) = (Q(z + \partial)q)\psi_z$ for every polynomial q, so the derivatives of $q\,\psi_z$ stay in the finite-dimensional space of polynomials of degree $\leq \deg q$ times ψ_z. Moreover $\prod_k \mathfrak{m}_{z_k}^{\deg q_k + 1} \subseteq \operatorname{Ann}(w)$, so $V(\operatorname{Ann} w) \subseteq \{z_k\}$; conversely $Q \in \operatorname{Ann}(w)$ forces $Q(z_k + \partial)q_k = 0$ by the linear independence of distinct exponentials, whose top-degree part is $Q(z_k)$ times that of q_k, so $Q(z_k) = 0$.

($\Leftarrow$) Let $\mathcal{R}w$ be finite-dimensional and p, q the minimal polynomials of ∂_t and ∂_x restricted to $\mathcal{R}w$, so that $p(\partial_t)w = 0$ and $q(\partial_x)w = 0$. On an open rectangle $I' \times J' \subset \Omega$, induction on $\deg p$ shows that every distribution with $p(\partial_t)v = 0$ is $\sum_{j,l} c_{jl}(x)\, t^l e^{\lambda_j t}$ with $c_{jl} \in \mathcal{D}'(J')$ and λ_j the roots of p (the base case uses that a distribution with $\partial_t V = 0$ on a rectangle is $c(x) \otimes 1$). Applying $q(\partial_x)$ and testing against functions dual to $t^l e^{\lambda_j t}$ gives $q(\partial_x)c_{jl} = 0$, so each c_{jl} is an exponential-polynomial of x. Thus w is an exponential-polynomial E near every point. The set where w coincides near y with a fixed E is open, and it is closed because two exponential-polynomials that agree on an open set agree everywhere; by connectedness it is Ω. □

A.2 Identification

Proof of Theorem 5.1. Let $I = \operatorname{Ann}(u) \ni P$ and write $I = G\,J$ as in Lemma 2.4. Since $P \in I$, $G \mid P$, so $G = c' \prod_i P_i^{f_i}$ with $0 \leq f_i \leq e_i$. First, $G \sim P$ iff $I = (P)$: if $G \sim P$ then $I = PJ \subseteq (P) \subseteq I$; if $I = (P)$, P divides all of I and lies in I.

(a), $\Leftarrow$ by contraposition. If $I \neq (P)$, some $f_i < e_i$, so $P/P_i = G\,H$ with $H \in \mathcal{R}$. A power $\mathfrak{n}^\ell$ of the ideal $\mathfrak{n}$ of the finite set $V(J)$ lies in J, and for $h \in \mathfrak{n}^\ell$, $h \cdot P/P_i = G(Hh) \in GJ = I$, i.e. $h(\partial)v_i = 0$. So $V(\operatorname{Ann} v_i) \subseteq V(J)$ is finite and $v_i \in \mathrm{EP}$ by Lemma 2.7.

(a), $\Rightarrow$ by contraposition. If $v_i \in \mathrm{EP}$, $V(\operatorname{Ann} v_i)$ is finite while $V(P_i)$ is infinite, so some $h \in \operatorname{Ann}(v_i)$ is not divisible by P_i. Then $Q = h \cdot P/P_i$ satisfies $Q(\partial)u = h(\partial)v_i = 0$; if $Q = PA$, cancelling P/P_i gives $h = P_i A$, a contradiction. So $Q \in I \setminus (P)$.

(b) $V(I) = V(G) \cup V(J)$ with $V(J)$ finite. If $f_i \geq 1$, $V(P_i) \subseteq V(I)$; if $f_i = 0$, $V(P_i) \cap V(I)$ lies in the finite set $\bigcup_{f_j \geq 1} V(P_i, P_j) \cup V(J)$, so $V(P_i) \not\subseteq V(I)$. Hence $V(I) = V(P)$ iff all $f_i \geq 1$, and $f_i = 0$ iff $v_i' \in \mathrm{EP}$ by the argument of (a) with $P/P_i^{e_i}$ in place of P/P_i.

(c) If $e_i = 1$ then $v_i = v_i'$. If P is monic of degree m in τ, divide any $Q \in \mathcal{R}$ by P in $\mathbb{C}[\xi][\tau]$: $Q = AP + \sum_{j<m} \varphi_j(\xi)\tau^j$ uniquely. Then $Q \in I$ iff $\sum_j \varphi_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $\varphi_j = 0$. □

Proof of Proposition 5.2. (a) If $I = (P)$, every annihilator is SP and $\deg SP \geq \deg P$ with equality iff S is a nonzero constant. (b) If $I = (P)$ then $G \sim P$ and $J = \mathcal{R}$, so $\Sigma_{\mathrm{p}}(u) = \emptyset$. Conversely, $V(J) = \emptyset$ gives $J = \mathcal{R}$ by the Nullstellensatz, hence $I = (G) = (P)$. (c) Let f_i be the multiplicity of P_i in G and $w = (P/P_i^k)(\partial)u$. If $k \leq e_i - f_i$, then $G \mid P/P_i^k$, say $P/P_i^k = GH$; for h in a power $\mathfrak{n}^\ell \subseteq J$ of the ideal of $V(J)$, $h \cdot P/P_i^k = G(Hh) \in I$, so $\mathrm{Ann}(w) \supseteq \mathfrak{n}^\ell$ and $w \in \mathrm{EP}$. If $k > e_i - f_i$ and $w \in \mathrm{EP}$, pick $h \in \mathrm{Ann}(w)$ with $P_i \nmid h$ (as in the proof of Theorem 5.1); then $hP/P_i^k \in I$, so $G \mid hP/P_i^k$, but P_i occurs in hP/P_i^k with multiplicity $e_i - k < f_i$, a contradiction. Hence $f_i = e_i - \max\{k : \ w \in \mathrm{EP}\}$. □

A.3 Decomposition and stability

Proof of Theorem 3.1. We prove the statement for open connected Ω with $H^1_{dR}(\Omega) = 0$, which holds for simply connected domains.

Step 1: finite modules have no obstruction. Let M be an $\mathcal{R}$-module of finite dimension over $\mathbb{C}$. Then $\mathrm{Ext}^1_{\mathcal{R}}(M, \mathcal{A}) = 0$ and $\dim \mathrm{Hom}_{\mathcal{R}}(M, \mathcal{A}) = \dim M$. By the long exact sequence it is enough to check one-dimensional modules $\mathcal{R}/\mathfrak{m}_z$, since M has a composition series with such quotients. A free resolution of $\mathcal{R}/\mathfrak{m}_z$ is the Koszul complex of $(\tau - \tau_z, \xi - \xi_z)$, so $\mathrm{Ext}^\bullet(\mathcal{R}/\mathfrak{m}_z, \mathcal{A})$ is the cohomology of $\mathcal{A} \to \mathcal{A}^2 \to \mathcal{A}$, $v \mapsto \big((\partial_t - \tau_z)v, (\partial_x - \xi_z)v\big)$, $(a, b) \mapsto (\partial_t - \tau_z)b - (\partial_x - \xi_z)a$. Multiplication by ψ_{-z} turns it into the de Rham complex of Ω with coefficients in $\mathcal{A}$, whose first cohomology is $H^1_{dR}(\Omega) = 0$ (for distributions by de Rham's theorem), and whose zeroth cohomology is the constants, of dimension one.

Step 2: two factors. For coprime Q_1, Q_2, $(Q_1) \cap (Q_2) = (Q_1 Q_2)$ by unique factorization, and $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ is exact ($1 \mapsto (1, 1)$, $(a, b) \mapsto a - b$). Applying $\mathrm{Hom}_{\mathcal{R}}(\cdot, \mathcal{A})$, and using $\mathrm{Hom}(\mathcal{R}/(f), \mathcal{A}) = \ker_{\mathcal{A}} f$, gives the exact sequence $0 \to \mathrm{Hom}(\mathcal{R}/(Q_1, Q_2), \mathcal{A}) \to \ker Q_1 \oplus \ker Q_2 \xrightarrow{(u_1,u_2)\mapsto u_1+u_2} \ker Q_1 Q_2 \to \mathrm{Ext}^1(\mathcal{R}/(Q_1, Q_2), \mathcal{A})$. The quotient $\mathcal{R}/(Q_1, Q_2)$ is finite-dimensional by Lemma 2.4, so by Step 1 the last group vanishes: the sum map is onto, and its kernel has dimension $\dim \mathcal{R}/(Q_1, Q_2)$ and consists of solutions of a zero-dimensional ideal, i.e. of exponential-polynomials.

Step 3: induction. $P' = Q_2 \cdots Q_N$ is coprime to Q_1, so $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j \geq 2} \ker Q_j$. Dimensions of kernels add along the composition of the two sum maps, and $\dim \mathcal{R}/(Q_1, P') = \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j)$ because $\mathcal{R}/(Q_1, Q_2Q_3 \cdots)$ is the direct sum of its localizations at the finitely many common zeros, where the intersection multiplicities add. This gives (7). □

Proof of Theorem 3.4. The sum operator $\mathcal{S}$ is bounded from $\bigoplus_j M_j$ into $\ker_{H^s} P$, and every u in its range has components in H^s. Its range is all of $\ker_{H^s} P$ iff the range is closed and dense. By the closed range theorem the range is closed iff $\mathcal{S}$ is bounded below on $(\ker \mathcal{S})^\perp$, i.e. iff $\gamma_N > 0$, and then the minimum-norm preimage satisfies (10). For two closed subspaces, $M_1 + M_2$ is closed iff $c_F < 1$, and $\gamma_2 = 1 - c_F$ [2]. In finite dimension every subspace sum is closed, and the sum equals the span of the sampled exponentials of P, each of which lies in the kernel of one factor. □

Proof of Proposition 4.2. For real $\zeta > c/\nu$ let $u_\zeta = e^{\lambda_\zeta t+\zeta x}$, $\lambda_\zeta = \nu\zeta^2 - c\zeta > 0$; then $Q(\partial)u_\zeta = 0$ and $\|u_\zeta\|_{H^s(\Omega)} \le C_s(1+\zeta)^{2s}e^{\lambda_\zeta T}\|e^{\zeta x}\|_{L^2(a,b)}$. With base time t_0 and horizon t, Theorem 4.1 gives

$$E_{\text{int}} = e^{\lambda_\zeta t_0}\big(e^{-c\zeta t}-1\big)\big(e^{\nu\zeta^2 t}-1\big)\,e^{\zeta x},$$

whose $L^2(\Omega)$ norm is at least a constant times $e^{\nu\zeta^2 t}\|e^{\zeta x}\|_{L^2(a,b)}$ for large ζ. With $t_0 = 0$ and $t = T$ the ratio therefore grows at least like

$$e^{\nu\zeta^2 T-\lambda_\zeta T}(1+\zeta)^{-2s} = e^{c\zeta T}(1+\zeta)^{-2s},$$

which is unbounded for every s. For the second statement, $v + w$ is killed by Q and by $R = (\tau + c\xi)(\tau - \nu\xi^2)$. At a common zero of Q and R, either $\tau = -c\xi$, and then $Q = 0$ gives $\nu\xi^2 = 0$, or $\tau = \nu\xi^2$, and then $c\xi = 0$; in both cases $(\tau, \xi) = (0, 0)$. So (Q, R) has a single zero, and its common kernel is a finite-dimensional space of exponential-polynomials (Lemma 2.7). □

B Primer for readers new to the concepts

This appendix *adds* concrete explanations; nothing in the main text depends on reading it. It covers four notions in this order: ideals and annihilators (§B.1), exponential-polynomials (§B.2), the Chinese remainder theorem (§B.3), and the companion matrix on a numerical example (§B.4).

B.1 Ideals, annihilators and the gcd, starting from integers

Integers. The multiples of 6 form an *ideal* $(6) = \{\dots, -6, 0, 6, 12, \dots\}$: it is closed under addition and under multiplication by any integer. The ideal generated by 4 and 6 is $(4, 6) = (2)$, since $2 = 6 - 4$ and every combination is even: the ideal "remembers" only the gcd.

Operators. Replace integers by operators and "multiple of 6" by "kills u". For $u = e^x$ every operator $Q(\partial_x)$ with $Q(1) = 0$ kills u, i.e. every multiple of $\partial_x - 1$. So

$$\begin{aligned}\operatorname{Ann}(e^x) &= (\partial_x - 1),\\ \operatorname{Ann}(\sin x) &= (\partial_x^2 + 1),\\ \operatorname{Ann}(xe^x) &= ((\partial_x - 1)^2),\\ \operatorname{Ann}(e^x + e^{2x}) &= ((\partial_x - 1)(\partial_x - 2)).\end{aligned}$$

In one variable every annihilator ideal has a single generator (the *minimal annihilator*), exactly as every ideal of integers is (n). In two variables an ideal can need several generators, as $(\tau - \tau_0, \xi - \xi_0)$ does for $u = e^{\tau_0 t+\xi_0 x}$.

The gcd. $\operatorname{Ann}(u) = G_u J_u$ separates the part of the ideal that is a single polynomial multiple (G_u, "infinite part") from the part J_u that has only finitely many common zeros (J_u, "finite debris"). Example: for $u = f(x+t)$ with f generic (not an exponential-polynomial), $\operatorname{Ann}(u) = (\partial_t - \partial_x)$ by Theorem 5.1, so $G_u = \tau - \xi$ and $J_u = \mathcal{R}$; for $u = e^{x+t}$, $\operatorname{Ann}(u) = (\tau - 1, \xi - 1)$, so $G_u = 1$ and $J_u = \operatorname{Ann}(u)$.

B.2 Exponential-polynomials: examples and a test

Function	In EP?	Reason (derivatives of u span ...)
xe^x	yes	span$\{xe^x, e^x\}$, dimension 2
$x^2e^{3t} + \cos x$	yes	finite sum of polynomial $\times$ exponential ($\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$)
e^{-x^2} (Gaussian)	no	$\frac{d^k}{dx^k} e^{-x^2} =$ (polynomial of degree k) e^{-x^2}: independent for all k, infinite dimension
$1/(1+x^2)$	no	derivatives are rational functions with growing denominators: infinite dimension

This is Lemma 2.7 in action: "$u \in$ EP" means "its derivatives span a finite-dimensional space".

B.3 The Chinese remainder theorem, concretely

Integers. The system $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 3$ has the unique solution $x \equiv 5 \pmod 6$: $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$. Coprimality of 2 and 3 is what makes the pair of remainders determine x.

One-variable ODE. Let $p(\partial) = (\partial - 1)(\partial - 2)$. Then $\ker p = \ker(\partial - 1) \oplus \ker(\partial - 2) =$ span$\{e^x, e^{2x}\}$, and the Lagrange-type projectors are explicit: if $u = ae^x + be^{2x}$,

$$u_1 = \frac{\partial - 2}{1 - 2}\, u = \; a\, e^x,$$
$$u_2 = \frac{\partial - 1}{2 - 1}\, u = \; b\, e^{2x}.$$

This is Eq. (11) with ∂_t replaced by ∂.

Two variables. For the wave equation $Q_1 = \tau - \xi$, $Q_2 = \tau + \xi$ are coprime, but the ideal $(Q_1, Q_2) = (\tau, \xi)$ is not the unit ideal: the two lines meet at the origin. The quotient $\mathcal{R}/(\tau, \xi) = \mathbb{C}$ has dimension 1, which is the constant ambiguity $\delta(P) = 1$ of the wave equation. Theorem 3.1 says that this finite overlap is the *only* failure of the integer picture.

B.4 The companion matrix on a numerical example

For $u'' - 3u' + 2u = 0$ the companion matrix is $C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, with characteristic polynomial $\lambda^2 - 3\lambda + 2$ and eigenvalues $1, 2$. The projectors of Eq. (4) are $\Pi_1 = \dfrac{C - 2\mathrm{Id}}{1 - 2}$ and $\Pi_2 = \dfrac{C - \mathrm{Id}}{2 - 1}$, and $e^{xC} = e^x \Pi_1 + e^{2x} \Pi_2$.

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